TabuLLM: AI Benchmark Report

1. Executive Summary: Consolidated Audit Results

This report consolidates the results of the TabuLLM data augmentation pipeline. The objective is to audit the fidelity, diversity, and privacy of the synthetic data generated via LLM inference. Below are the averaged performance indicators across all processed datasets in this session.

Audit Indicator	Global Session Mean (± Std)
Synthetic Fidelity (JSD)	75.9 ± 3.30
Correlation Stability	90.3 ± 1.10
Distribution Consistency	86.0 ± 3.03
Entropy Preservation	48.1 ± 1.04
Mean Z Score Drift	0.3 ± 0.03
Categorical Fidelity	65.3 ± 3.07
Privacy Risk	2.7 ± 1.00
Data Density	100.0 ± 0.00
Coverage Score	59.3 ± 4.15
Feature Accuracy	69.2 ± 1.47
Novelty Score	0.9 ± 0.01

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Dataset Audit: indian_liver_patient_dataset.csv

1. Dataset Portrait

Attribute	Value
Rows	583
Columns	11
Numeric features	10
Categorical features	1
Missing cells (%)	0.1%

2. Generation Strategy & LLM Config [1]

Parameter	Configuration
Algorithm (*12)	google/gemma-4-e2b
Folds (*13)	10
Context % (*14)	0%
Generation % (*15)	100%
Join context (*16)	False

3. Statistical Significance & Data Drift

Feature	Orig Mean	Aug Mean	Shift	P-Value	Status
Age	44.746	40.485	-4.261	9.5%	OK
TB	3.299	1.698	-1.601	48.5%	Drift
DB	1.486	0.489	-0.997	67.1%	Drift
Alkphos	290.576	191.112	-99.465	34.2%	Drift
Sgpt	80.714	32.845	-47.868	59.3%	Drift
Sgot	109.911	39.549	-70.361	64.0%	Drift
TP	6.483	6.080	-0.403	6.2%	OK
ALB	3.142	2.881	-0.261	8.3%	OK
A/G Ratio	0.947	0.914	-0.033	3.5%	OK
Selector	1.286	1.103	-0.183	14.3%	Warn

4. Quality & Fidelity Metrics (Mean ± Std)

Metric Category	Aggregated Score
Synthetic Fidelity (JSD) (*1)	75.9 ± 3.30
Correlation Stability (*2)	90.3 ± 1.10
Distribution Consistency (*3)	86.0 ± 3.03
Entropy Preservation (*4)	48.1 ± 1.04
Mean Z Score Drift (*5)	0.3 ± 0.03
Categorical Fidelity (*6)	65.3 ± 3.07
Privacy Risk (*7)	2.7 ± 1.00
Data Density (*8)	100.0 ± 0.00
Coverage Score (*9)	59.3 ± 4.15

Feature Accuracy (*10)	69.2 ± 1.47
Novelty Score (*11)	0.9 ± 0.01

[1] (*12) Algorithm: Contextual constraint or model parameter.

[2] (*13) Folds: Contextual constraint or model parameter.

[3] (*14) Context %: Contextual constraint or model parameter.

[4] (*15) Generation %: Contextual constraint or model parameter.

[5] (*16) Join context: Contextual constraint or model parameter.

[6] Synthetic Fidelity (JSD): Overall distribution similarity.
[7] Correlation Stability: Multivariate relationships preserved.
[8] Distribution Consistency: Median feature similarity.
[9] Entropy Preservation: Information diversity maintained.
[10] Mean Z Score Drift: Standardized distribution shift.
[11] Categorical Fidelity: Category distribution similarity.
[12] Privacy Risk: Synthetic rows too close to originals.
[13] Data Density: Non-missing cells in augmented data.
[14] Coverage Score: Synthetic coverage of original space.
[15] Feature Accuracy: Mean feature similarity.
[16] Novelty Score: Distance ratio to nearest neighbors.

6. Downstream ML Utility: Comparative Analysis

Model: LR

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.7059829059829059	0.7076923076923076	0.7094017094017093	0.0034188034188034288	
F1	0.8128002697439015	0.8277001433368552	0.8242587226584037	0.011458452914502325	

Model: DT

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.6641025641025642	0.6991452991452991	0.6854700854700855	0.021367521367521347	
F1	0.7663032365208943	0.8116224397898806	0.7896951512462479	0.023391914725353624	

Model: XGB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.7025641025641025	0.7128205128205127	0.7205128205128205	0.01794871794871795	
F1	0.7997903571549948	0.8292984176412984	0.8163404652574296	0.016550108102434846	

Model: GB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.6854700854700855	0.7068376068376068	0.7145299145299145	0.029059829059829078	
F1	0.7909408486522184	0.8255516940019823	0.8166991668215523	0.025758318169333883	*

Model: AB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.6743589743589744	0.7119658119658119	0.6871794871794872	0.01282051282051282	
F1	0.7821343746683586	0.8284598600934243	0.8044941341143298	0.022359759445971244	

Model: KNN

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Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.6863247863247863	0.7068376068376068	0.7128205128205127	0.026495726495726478	
F1	0.7971299319768493	0.8279091624529537	0.8277803606086902	0.030650428631840743	*

Model: NB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.558974358974359	0.6803418803418804	0.5923076923076923	0.03333333333333334	*
F1	0.5587432119380917	0.7725278294746187	0.6138910848456949	0.05514787290760312 5	*

Model: MLP

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.6897435897435897	0.6786324786324787	0.6982905982905983	0.00854700854700854	
F1	0.8011016273174306	0.8062418960323852	0.8171894932723027	0.01608786595487215	

Model: SVM

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.7068376068376068	0.7068376068376068	0.7068376068376068	0	
F1	0.8279091624529537	0.8279091624529537	0.8279091624529537	0	

Model: RF

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.7145299145299145	0.7059829059829059	0.7205128205128205	0.005982905982905984	
F1	0.8154972140649285	0.8261528029988279	0.8294180643531904	0.013920850288261922	

Appendix: Benchmark Data Trace

Legend: [Blue] = Real Context | [Green] = Generated Synthetic | [Yellow] = Real Training (Remaining) | [Grey] = Real Test Baseline

A. Baseline: Original Dataset (Full Record Sample)

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	65	Female	0.7	0.1	187	16	18	6.8	3.3	0.9
2	62	Male	10.9	5.5	699	64	100	7.5	3.2	0.74
3	62	Male	7.3	4.1	490	60	68	7.0	3.3	0.89
4	58	Male	1.0	0.4	182	14	20	6.8	3.4	1.0
5	72	Male	3.9	2.0	195	27	59	7.3	2.4	0.4
6	46	Male	1.8	0.7	208	19	14	7.6	4.4	1.3
7	26	Female	0.9	0.2	154	16	12	7.0	3.5	1.0
8	29	Female	0.9	0.3	202	14	11	6.7	3.6	1.1
9	17	Male	0.9	0.3	202	22	19	7.4	4.1	1.2
10	55	Male	0.7	0.2	290	53	58	6.8	3.4	1.0
11	57	Male	0.6	0.1	210	51	59	5.9	2.7	0.8
12	72	Male	2.7	1.3	260	31	56	7.4	3.0	0.6
13	64	Male	0.9	0.3	310	61	58	7.0	3.4	0.9
14	74	Female	1.1	0.4	214	22	30	8.1	4.1	1.0
15	61	Male	0.7	0.2	145	53	41	5.8	2.7	0.87
16	25	Male	0.6	0.1	183	91	53	5.5	2.3	0.7
17	38	Male	1.8	0.8	342	168	441	7.6	4.4	1.3
18	33	Male	1.6	0.5	165	15	23	7.3	3.5	0.92
19	40	Female	0.9	0.3	293	232	245	6.8	3.1	0.8
20	40	Female	0.9	0.3	293	232	245	6.8	3.1	0.8

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B.1 Audit Trace: Fold 1 | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	56	Male	1.4506897941286678	0.1108741715398966	196	22	44	5.110755748339753	2.058077028671483	0.5485075422861231
2	26	Male	0.8665887466008302	0.041636460116106	197	40	28	5.83215970463974	2.2362077627226298	0.5617037230978511
3	63	Male	2.199160564206201	0.2965357867191053 5	243	50	68	6.0190944733566	2.1881816098151243	0.5047841605779495
4	55	Male	1.3162552044039368	0.5337313630045202	212	45	50	6.151694447615361	2.4336941986834253	0.7559751465539207
5	45	Male	0.4674079372121127	0.3752760380881729	119	37	68	7.2503089273952954	2.7940050707995683	1.0291354536058621
6	59	Male	1.0469717811182177	0.1408743688831189	198	14	27	5.720097643863168	1.65504621082456	0.6901636054475402
7	33	Male	7.1	3.7	196	622	497	6.9	3.6	1.09
8	38	Male	2.2	1.0	310	119	42	7.9	4.1	1.0
9	64	Male	3.0	1.4	248	46	40	6.5	3.2	0.9
10	40	Female	0.9	0.2	285	32	27	7.7	3.5	0.8

B.2 Audit Trace: Fold 2 | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	19	Male	1.6469294293583512	0.4418678194655075 7	146	39	53	7.1209781322194505	4.7384414740536105	0.8466622141471339
2	48	Male	1.4051489591810902	-0.12717999847089378	167	46	70	6.615496757222395	2.5094500609114774	0.629015200651 99
3	50	Male	1.2447150340072956	0.2348626337764725	83	19	35	7.190992432842653	3.2266737455370587	0.8299039990246712
4	24	Male	1.5132635033637911	1.9329279701429117	227	35	85	7.008268471956504	2.0926745711074353	0.7942328936421729
5	50	Male	0.5110537742034571	0.0159882064842642 45	149	6	36	7.168698866802985	3.038421806378432	0.9379091645731363
6	24	Male	4.7954029244335354	1.6046172379609256	364	39	61	5.772451825268417	2.7876057092375497	0.4703048725547231
7	50	Female	1.0	0.3	191	22	31	7.8	4.0	1.0
8	52	Female	0.6	0.1	194	10	12	6.9	3.3	0.9
9	30	Male	0.8	0.2	182	46	57	7.8	4.3	1.2
10	58	Male	0.8	0.2	298	33	59	6.2	3.1	1.0

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B.3 Audit Trace: Fold 3 | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	28	Male	1.3275899916928258	0.5193837718526113	268	27	64	6.852589152290659	3.886673240704443	1.5108185349023242
2	43	Male	1.0746925455259855	0.13582905368016757	195	27	31	4.833536076171194	2.350838063344941	0.5015067133707504
3	29	Male	0.4797320336291026	0.2627732742954560	173	29	47	7.556805239475309	3.387646091454658	1.0840389975249731
4	40	Male	1.0655010874132336	0.3623704982979148	242	42	58	4.319916000407378	2.694184308731090	0.5366528510599625
5	32	Male	0.5810638597790398	0.5165848511589283	111	16	-1	6.529312670939582	3.625542578568027	1.3297854045194732
6	15	Male	1.3099482128359732	0.4190484109422857	207	43	89	6.398223506669456	3.203776387309416	0.7044047586308938
7	44	Female	1.9	0.6	298	378	602	6.6	3.3	1.0
8	30	Male	0.8	0.2	182	46	57	7.8	4.3	1.2
9	53	Male	0.9	0.4	238	17	14	6.6	2.9	0.8
10	28	Male	0.8	0.3	190	20	14	4.1	2.4	1.4

B.4 Audit Trace: Fold 4 | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	54	Male	0.6702674013254186	-0.12026902219402147	187	15	47	6.64465742908024	3.762414324493943	1.5891804506638991
2	62	Male	1.2399211464698965	0.4652850515290674 6	144	50	58	6.980566921618799	4.016755473890894	1.6661733115073868
3	62	Male	0.7197747533936727	0.1981132429415133 3	100	24	-6	6.772670692456935	2.921336689559116	1.0855734534898125
4	57	Male	1.5607404379859868	0.8467071343268722	254	25	36	5.983154832387858	1.775274849567191 3	0.4130564035853005
5	32	Male	1.1610964686600285	0.2646798454688567	207	32	-13	6.922020860777811	4.006979878240124 5	1.3148195395143947
6	47	Male	1.69072670369656	0.12548870995095915	194	72	45	4.97942213575236	2.37154070628235	0.9542036043968065
7	42	Male	0.7	0.2	197	64	33	5.8	2.4	0.7
8	69	Male	0.9	0.2	215	32	24	6.9	3.0	0.7
9	53	Male	1.6	0.9	178	44	59	6.5	3.9	1.5
10	49	Male	1.3	0.4	206	30	25	6.0	3.1	1.06

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B.5 Audit Trace: Fold 5 | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	59	Male	1.2357250845607748	0.05767717518621884	188	38	51	4.788621463793376	1.3959985286257477	0.7188972718973676
2	37	Male	0.7665175244826992	0.32142770254649394	221	56	48	6.592936390265062	3.4761678935512257	1.0351440080378487
3	40	Male	1.1386707000155007	0.15057284753181377	192	1	23	7.868803874057857	3.1364895246995506	1.0830529631762666
4	42	Male	1.138655319090317	0.23955891298949258	237	41	25	6.600761739920617	4.0992439407151819	0.9491774866468478
5	28	Male	0.4495902665331043	0.4489205599985142	298	22	43	7.310068510874069	3.8904604265613356	0.7122334954915625
6	57	Male	1.6410843966167288	0.44520718114422125	148	25	70	6.661047808459481	3.8070185898397058	1.0090253892818477
7	33	Male	2.1	0.7	205	50	38	6.8	3.0	0.7
8	22	Female	6.7	3.2	850	154	248	6.2	2.8	0.8
9	48	Male	5.8	2.5	802	133	88	6.0	2.8	0.8
10	84	Female	0.7	0.2	188	13	21	6.0	3.2	1.1

B.6 Audit Trace: Fold 6 | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	49	Male	1.9965804491577435	0.5268598247632146	196	39	51	5.89071637347940 9	2.4405890831278727	0.6192118978601282
2	24	Male	1.5298343649329 09	0.7470208624518851	204	39	55	6.874214286648106	2.3390847519179983	0.8295551476025286
3	18	Male	1.2059749178881227	0.4174422717862042	190	40	34	7.347323440777214	3.9806144857574 72	1.5739163807132 72
4	50	Male	1.0845642089290806	0.16599067204484333	171	30	84	7.05812709099367	3.8809256503356018	0.7480599819174588
5	30	Male	1.0392253617071677	0.2335343234377 2096	156	46	39	7.400278452112934	3.9699981985741 18	0.8158658998266893
6	32	Male	1.994098129901679 3	-0.12812873721688	203	33	26	7.766556479447131	4.5585919771726 02	1.3407574132544753
7	50	Male	0.9	0.3	194	190	73	7.5	3.9	1.0
8	49	Male	0.6	0.1	218	50	53	5.0	2.4	0.9
9	18	Male	0.8	0.2	228	55	54	6.9	4.0	1.3
10	30	Female	0.7	0.2	194	32	36	7.5	3.6	0.92

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B.7 Audit Trace: Fold 7 | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	22	Male	4.48501561230974 7	5.55448846909626	238	38	21	7.48110336974672 3	3.56284619297685 27	1.26012254620798 75
2	60	Male	0.84710148303951 36	0.52068669937651 93	179	17	48	6.65719483053971 7	3.35759734534744 55	1.05155225193018 23
3	39	Male	0.63855435423805 05	0.08799201350556 543	157	41	14	6.93169470958605 95	3.85381658437100 73	1.12338207748307 43
4	57	Male	1.59619809647664 51	0.22813393246126 404	123	43	30	8.27014362480048 6	4.60470879976430 95	1.82499268732720 67
5	15	Male	0.57385049884200 41	0.24212269953575 463	198	38	27	7.57180621149029 3	3.88249211486228 2	1.16064652217575 76
6	61	Male	1.39023634194124 16	0.16577467639754 223	178	38	49	6.95336384442889 4	3.95062115265753 46	1.02700883124626 09
7	13	Female	0.7	0.2	350	17	24	7.4	4.0	1.1
8	54	Female	23.2	12.6	574	43	47	7.2	3.5	0.9
9	63	Male	0.9	0.2	194	52	45	6.0	3.9	1.85
10	46	Male	0.6	0.2	290	26	21	6.0	3.0	1.0

B.8 Audit Trace: Fold 8 | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	35	Male	0.40171822252324 485	1.30219375422194 7	191	39	35	6.14533464373506	2.22433834511117 33	0.4519079870444
2	39	Male	0.59512622450700 4	0.33597915954606 195	162	32	54	6.30591221777656 4	2.71863906579069 5	0.88783624216959 07
3	50	Male	0.72090167858130 76	0.22495267358768 503	191	22	88	6.12958741664575	2.25332832713226 95	0.79910800926730 21
4	31	Male	0.92363019120146 4	0.27327481489458 416	185	23	51	6.36540279726131 2	2.89384513365784 7	0.68560438188446 9
5	38	Male	0.95676309108002 53	0.20224003949994 232	182	19	55	6.15981742699116 9	4.47535602635023 5	1.00456028431396 67
6	32	Male	1.02411790005406 79	0.06290685974163 177	124	41	66	7.09827948037387 65	2.90515522143156 35	0.61933077094850 42
7	48	Male	0.7	0.2	208	15	30	4.6	2.1	0.8
8	60	Male	22.8	12.6	962	53	41	6.9	3.3	0.9
9	27	Male	1.0	0.3	180	56	111	6.8	3.9	1.85
10	28	Female	0.8	0.2	309	55	23	6.8	4.1	1.51

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B.9 Audit Trace: Fold 9 | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	59	Male	1.3792488282497712	0.4439726239393073	140	41	-5	6.750869235661836	2.8612570802404704	0.63833014654514
2	43	Male	1.3025280899979375	0.5492405807469676	213	33	40	5.433184674451687	3.6564096014533316	0.8481986302690079
3	53	Male	0.9002702290504379	0.2024383899447026	214	47	25	7.080695341806036	2.6096396713748025	0.7477875395923612
4	42	Male	2.101253693408937	1.3679933646833495	172	10	55	5.553362304339597	3.2892191719301602	0.6364988818293017
5	36	Male	1.0149160746958914	0.0531537214307411 24	127	28	39	8.094729538517583	4.0913977079232804	1.6843754735670464
6	34	Male	0.8811183602422901	-0.06839085667105405	195	17	35	7.618373168950614 4	3.1107120702773297	0.8435211843975986
7	26	Male	2.0	0.9	195	24	65	7.8	4.3	1.2
8	47	Male	2.7	1.3	275	123	73	6.2	3.3	1.1
9	74	Male	1.0	0.3	175	30	32	6.4	3.4	1.1
10	48	Male	3.2	1.6	257	33	116	5.7	2.2	0.62

B.10 Audit Trace: Fold 10 | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	42	Male	1.2893130629819258	-0.06833697603225114	93	30	54	7.206248665154316	3.9902970948390486	1.0882786247973766
2	32	Male	1.2735591606946883	0.1964565748290987	246	22	50	6.648995289723954	2.7902373937260331	1.1367007763704395
3	42	Male	1.0107060358476588	0.3310869931286912	148	34	26	6.028761585373974	3.8329644681280977	1.5579354653722037
4	31	Male	22.165517888070873	12.513655829635029	247	63	34	7.129154305609957 6	2.8384733730893933	0.6635135417200089
5	37	Male	0.8397690277173645	0.016280577677268093	150	45	14	7.262076781149327	4.1822989076628007	0.8981815614565096
6	63	Male	11.710432626491471	2.0481284374680602	195	20	60	5.119376177548134	2.5082826704167456	0.7377289949066197
7	54	Female	5.5	3.2	350	67	42	7.0	3.2	0.8
8	43	Male	22.5	11.8	143	22	143	6.6	2.1	0.46
9	57	Male	4.5	2.3	315	120	105	7.0	4.0	1.3
10	65	Male	0.7	0.2	199	19	22	6.3	3.6	1.3